# Lower Limb Auxiliary Exoskeleton with Gait Management Feedback for Hypotonia

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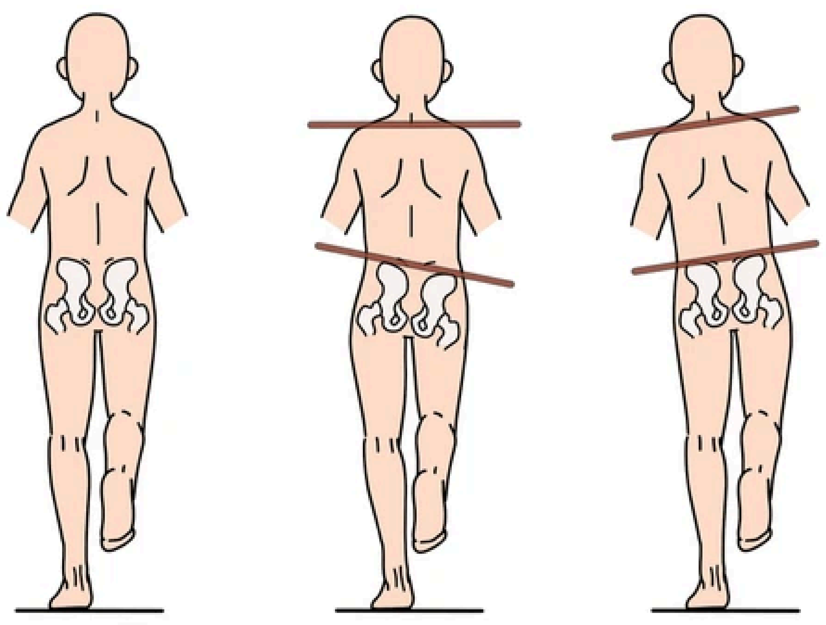
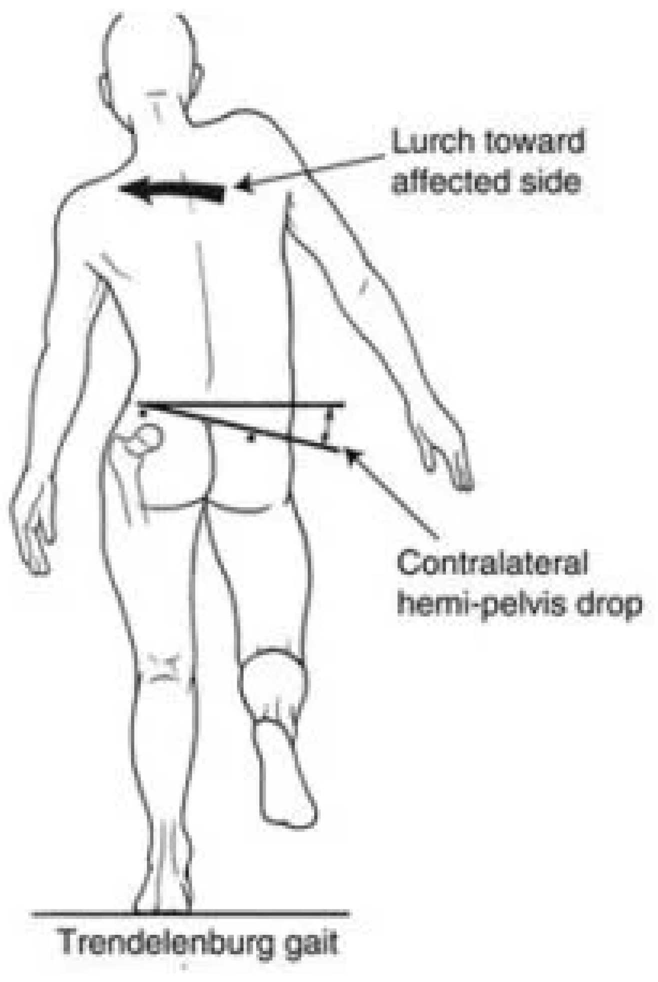
## PROBLEM

- Hypotonia: when a person's muscles are too weak or loose, making it harder to move and stay balanced
- Why is hypotonia such a significant issue?
  - Difficulty staying balanced can lead to difficulty navigating stairs, slopes, and slippery surfaces
    - Especially dangerous in Calgary, a hilly city where ice covers the ground for months
  - Long-term damages that hypotonia can cause include anterior pelvic tilt and increased vulnerability to bone fractures
- What is the scope of this issue?
  - Approximately 50% of children with Autism Spectrum Disorder experience hypotonia
  - Estimation of 80,000 children with hypotonia presenting autism in Canada
    - Calculations: [(6,012,795/36,991,981)×1,000,000]×0.5=81,271
    - The exact incidence of hypotonia is difficult to determine because it is not a disease but a "symptom"
- What are the current solutions?
  - Physical therapy
    - Exercises to strengthen muscles
    - Braces for support when walking

    Don't address balance issues when walking
  - Nutritional support
  - Enzyme replacement therapy, gene therapy



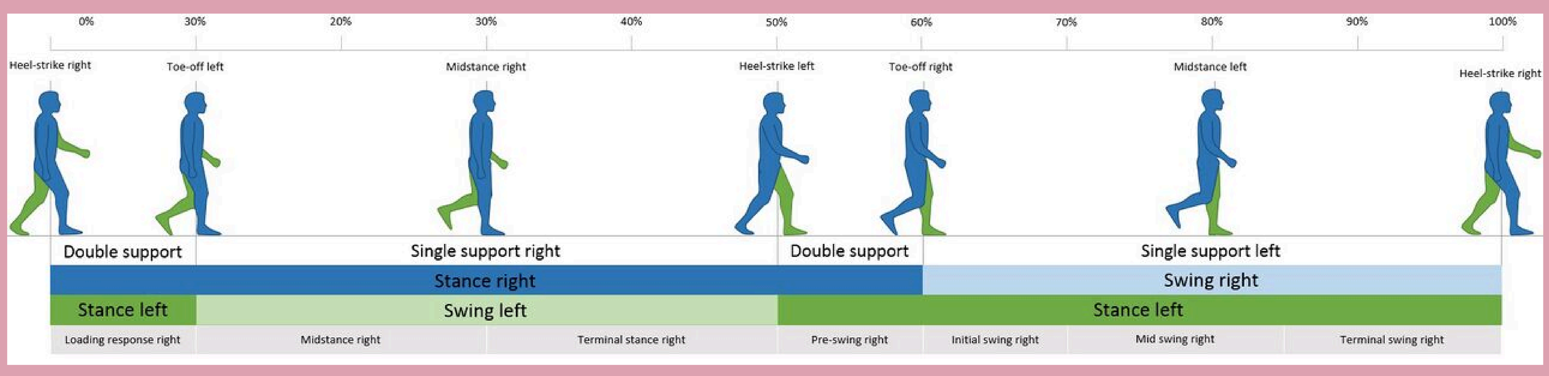



## METHOD

- This project aims to design an exoskeleton that 1) provides structural support to the lower limbs and 2) aide in developing a balanced walking gait—an aspect crucial for promoting musculoskeletal health. The second goal is significant because of the pivotal role that muscle engagement plays in fostering a balanced gait and preventing adverse outcomes such as poor posture and falling (Stotz et al., 2023).
- The design objectives of this walking assistance exoskeleton can be broken down into its mechanical design objectives and system control objectives.
  - Mechanical Design Objectives: The exoskeleton's frame is broken down into individual metal bars that act as the limbs, with rotational components that serve as the knee and hip joints. The rotational connection parts, powered by steering motors and gears, should have a reasonable degree of mobility corresponding to the user's movements. The torso attachment should support the spinal cord when wrapped around the user's waist.
  - System Control Objectives: The steering motors, connected to an Arduino Nano Board, should move the rotational components according to the user's range of mobility.

Maximum Range of Motion (ROM) at Lower Extremity Joints during the Gait Cycle

| Body part | Maximum ROM values |
|---|---|
| Hip | 20° of extension; 20° of flexion |
| Knee | 0° (complete extension); 60° of flexion |
| Ankle | 5° of dorsiflexion; 20° of plantarflexion |




Waist Support

Arduino Control Board

12V Battery

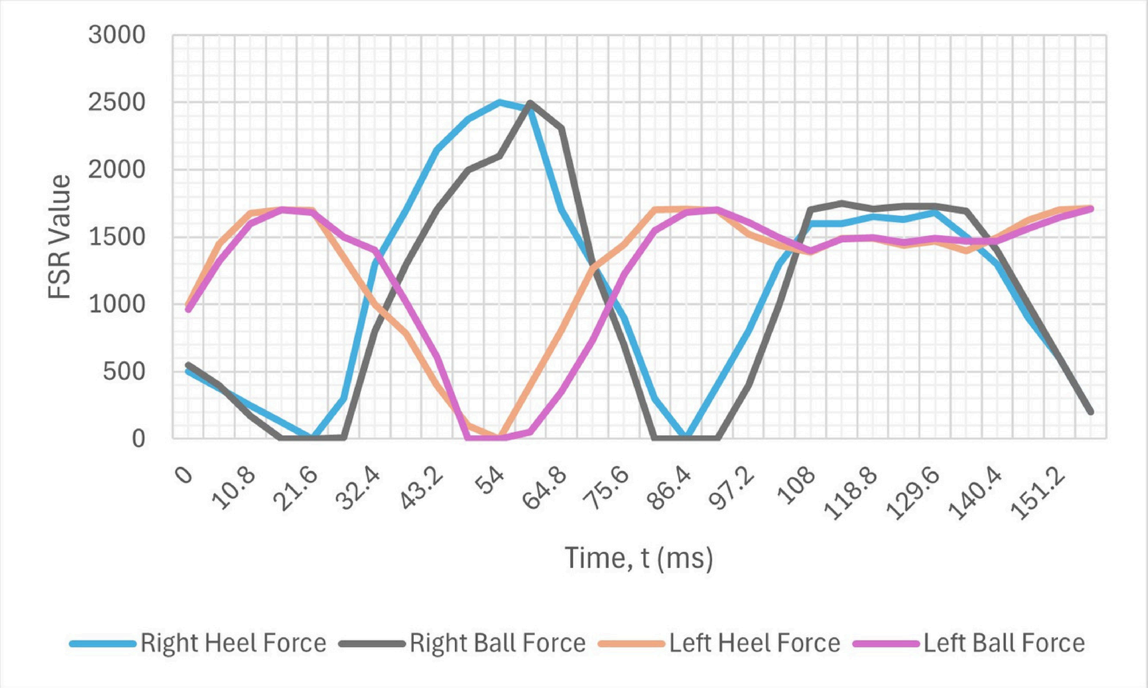
Hip Servo

Knee Servo

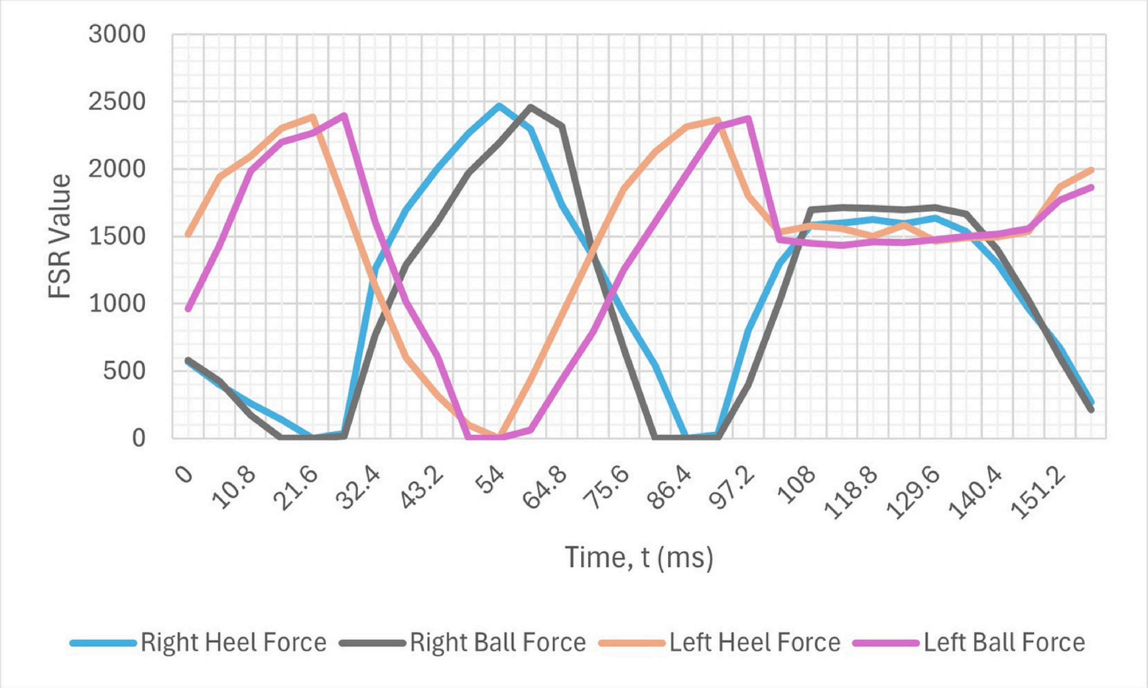
3D Printed Leg Attachment

# ANALYSIS

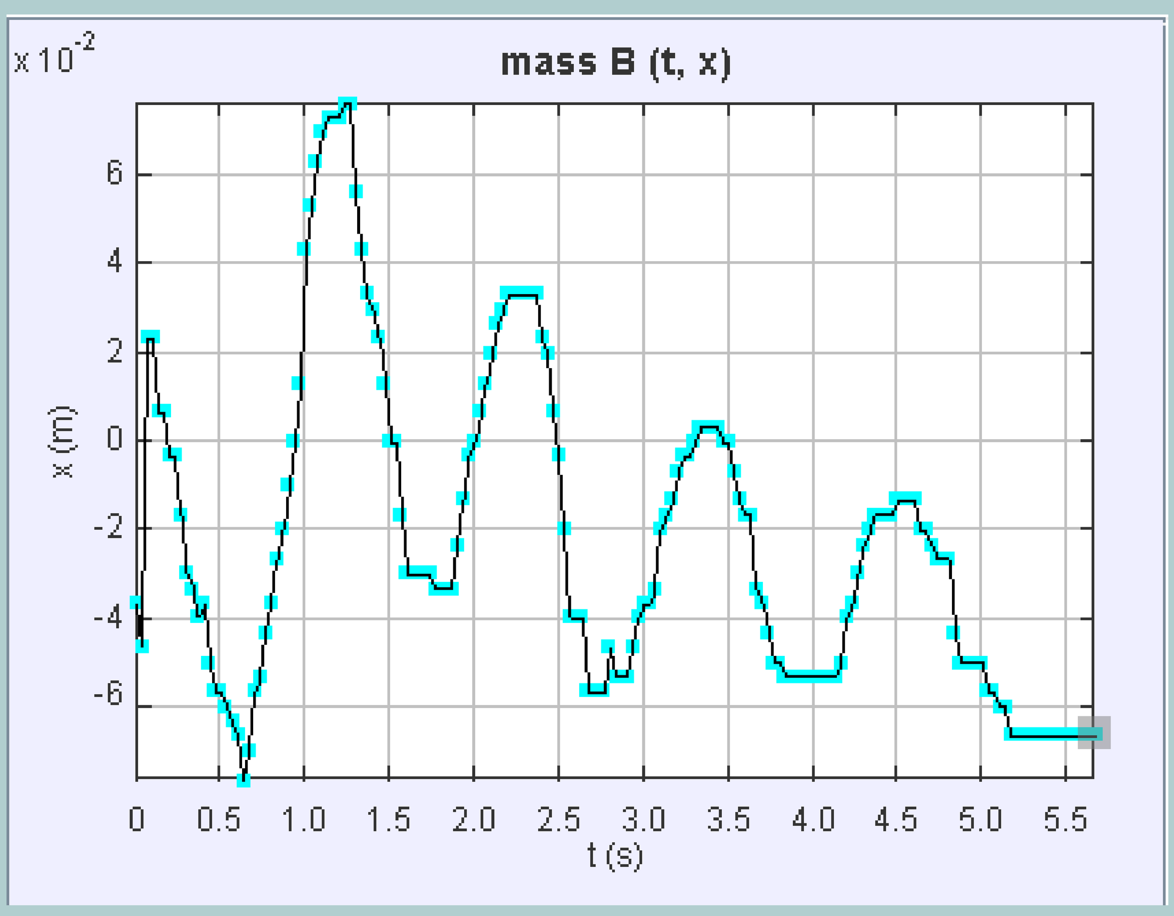
Graph 1. Pressure Detected by Heel and Ball FSR Sensors without Exoskeleton



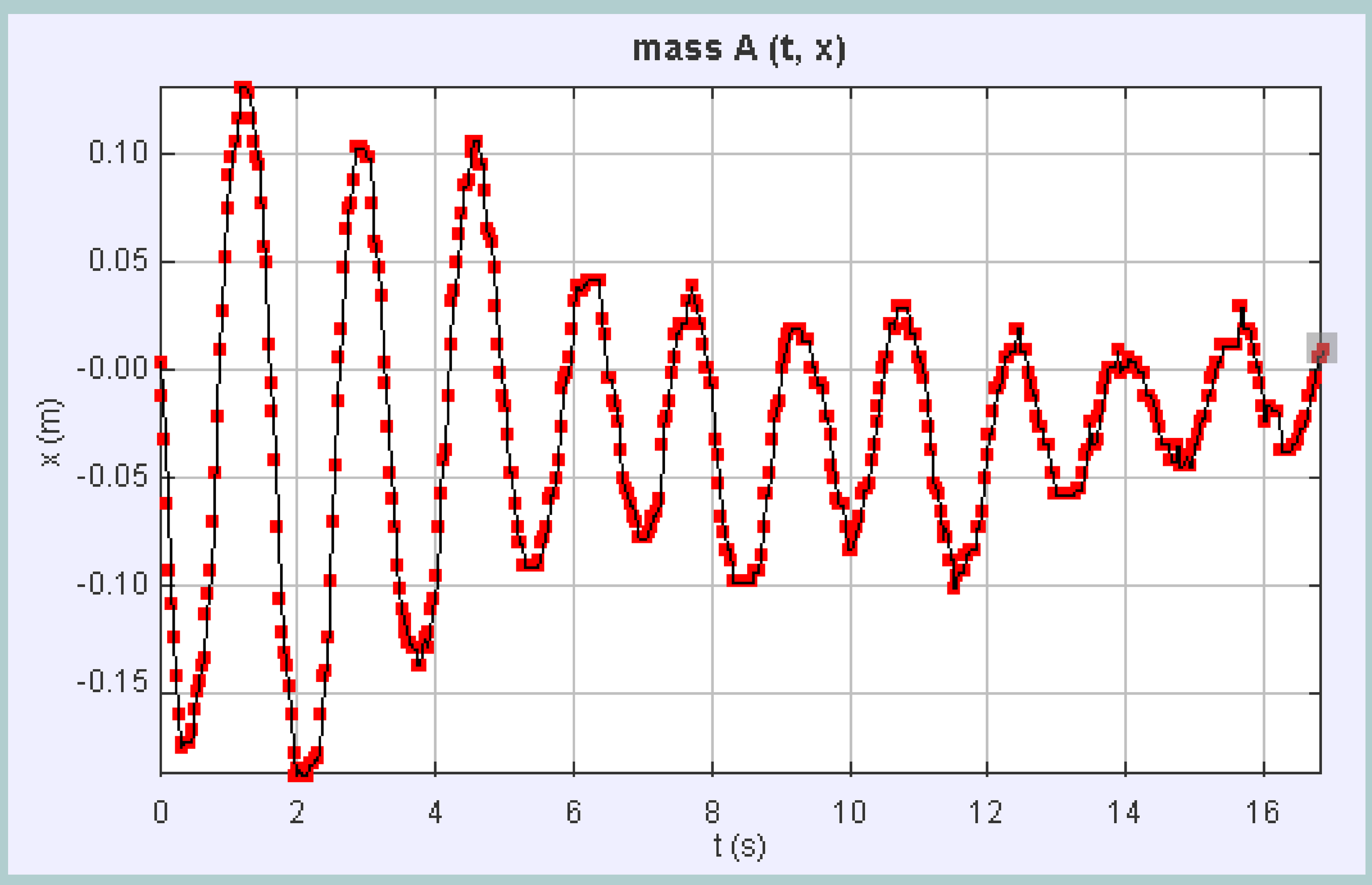

Graph 2. Pressure Detected by Heel and Ball FSR Sensors with Exoskeleton




Graph 3. Center of Mass Shift without Exoskeleton




Graph 4. Center of Mass Shift with Exoskeleton



- Graph 1
    - The pressure on her right foot is significantly higher than on her left foot
    - This discrepancy suggests that the participant's weight is mostly on her right leg, leading to an unbalanced walking gait
- Graph 2
    - Shows a more minor difference between the force exertion of each leg when the participant walks with the exoskeleton on
    - The difference between the maximum pressure values in Graph 2 is 89.8% less than that of Graph 1

Therefore, the pressure analysis concludes that the exoskeleton effectively balances the pressure on each foot for an individual with hypotonia.

- Graph 3
    - The imbalance of the participant's gait is especially apparent from seconds 1.0 to 2.0, where the center of mass shifts to the right of the origin significantly more than to the left (8 cm to the right, then 3 cm to the left)
    - Indicates that the participant is placing most of her weight on her right leg when walking.
- Graph 4
    - The right and left shifts are comparably more balanced
    - From seconds 3 to 5, the center of mass shift is roughly 10 cm to the right, 13 cm to the left, 11 cm to the right, and 9 cm to the left.
    - The maximum difference in shift here is 3 cm, which is lower than the difference of 5 cm in Graph 3

Therefore, the analysis of the center of mass shift concludes that the exoskeleton is effective at balancing out weight shift, leading to a more balanced walking gait.

# CONCLUSION

- Goal: Test if a lower limb exoskeleton can help a person with hypotonia walk more evenly and safely.
- Results:
    - Balanced weight between both feet when walking with the exoskeleton.
    - Smaller shifts in the body’s center of mass (less swaying side to side).
    - Improved stability and a more balanced walking pattern overall.
- Key Takeaway:
    - The exoskeleton could be a useful physiotherapy tool for people with hypotonia—not just for support, but also to improve their balance when walking.
- Why This Matters:
    - Unlike regular braces that only support the legs, this exoskeleton actively helps correct unbalanced gait, which may reduce future problems like joint pain or poor posture.
- Challenges Faced:
    - Difficulties in programming the motors (Servos) to mimic natural leg movement.
    - Lots of trial and error to get the walking motion right.
- Next Steps:
    - Make it easier to walk up/down stairs and on uneven ground.
    - Improve the system to help with waddling gait, which was still noticeable.
    - Waddling gait = swaying side to side like a duck.
        - Caused by weakness in the hip abductor muscles.
        - Future updates will focus on reducing this swaying.
- Potential Impact:
    - With more testing and improvements, this exoskeleton could be used by people with other mobility challenges, not just hypotonia.

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